

Definition of the Function

We define the function

$$f(x, y) = \frac{1}{1 + \left\| C \begin{bmatrix} x \\ y \end{bmatrix} \right\|^2},$$

where $C \in \mathbb{R}^{2 \times 2}$ is a transformation matrix controlling anisotropy.

Choice of Matrices C

The following matrices illustrate different anisotropies:

$$C_1 = \begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix} \quad (\text{vertical stretching}),$$

$$C_2 = \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix} \quad (\text{horizontal stretching}),$$

$$C_3 = \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix} R(30^\circ) \quad (\text{tilted ellipse}),$$

$$C_4 = 5I_2 \quad (\text{steeper dome}),$$

$$C_5 = 0.2I_2 \quad (\text{flatter dome}),$$

where I_2 denotes the 2×2 identity matrix.

Rotation Matrix

The general 2D rotation matrix is

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$$

For example, setting $\theta = 30^\circ$ yields

$$R(30^\circ) = \begin{bmatrix} \cos 30^\circ & -\sin 30^\circ \\ \sin 30^\circ & \cos 30^\circ \end{bmatrix}.$$